F18CF – Linear Algebra

Tutorial Sheet 1 – Solutions

1. (a) We apply Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & -1 & 2 & 2 \end{array}\right) \overset{\substack{R_2\to R_2-2R_1 \\ R_3\to R_3-R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 3 & -1 \\ 0 & -3 & 3 & 0 \end{array}\right) \overset{R_3\to R_3-R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 3 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

The last row corresponds to the equation $0x+0y+0z=1$ which has no solution. Hence the system is inconsistent.

(b) We apply Gaussian elimination

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 3 \\ 1 & 4 & 9 & 7 \end{array}\right) \overset{\substack{R_2\to R_2-R_1 \\ R_3\to R_3-R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 6 \end{array}\right) \overset{R_3\to R_3-3R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 0 \end{array}\right)$$

Every variable being a pivot variable, the system has a unique solution, and by substitution we get: $z=0$, $y=2$, $x=-1$.

(c) We apply Gaussian elimination

$$\left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 2 & -1 & -3 & 2 \\ 3 & 0 & -7 & 3 \end{array}\right) \overset{\substack{R_2\to R_2-2R_1 \\ R_3\to R_3-3R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 3 & -5 & 0 \\ 0 & 6 & -10 & 0 \end{array}\right) \overset{R_3\to R_3-2R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 3 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The system is thus consistent, and the only free variable is z. We get:

$z=\alpha\in\mathbb{R}$,

$3y-5z=0$, hence $y=\frac{5}{3}\alpha$, and

$x=2y-z+1=\frac{7}{3}\alpha+1$.

The general solution is of the form $(\frac{7}{3}\alpha+1,\frac{5}{3}\alpha,\alpha)$.

2. (a) We apply Gaussian elimination:

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 2 & 1 & 2 & -1 & 0 \\ 4 & 3 & 0 & 1 & 0 \\ 5 & 3 & 3 & -1 & 0 \end{array}\right) \overset{\substack{R_2\to R_2-2R_1 \\ R_3\to R_3-4R_1 \\ R_4\to R_4-5R_1}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 4 & -3 & 0 \\ 0 & -1 & 4 & -3 & 0 \\ 0 & -2 & 8 & -6 & 0 \end{array}\right) \overset{\substack{R_3\to R_3-R_2 \\ R_4\to R_4-2R_2}}{\rightsquigarrow}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 4 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The system is thus consistent, with free variables z and u. We thus set $u = \alpha$, $z = \beta$, with $\alpha, \beta \in \mathbb{R}$ and we get:

$y = 4\beta - 3\alpha$,

$x = -y + z - u = 2\alpha - 3\beta$.

(b) We apply Gaussian elimination:

$$\left(\begin{array}{cccc|c} 3 & 1 & 1 & 1 & 12 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1 & 12 \\ 1 & 1 & 1 & 3 & 0 \end{array}\right) \overset{\substack{R_2 \to 3R_2 - R_1 \\ R_3 \to 3R_3 - R_1 \\ R_4 \to 3R_4 - R_1}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 3 & 1 & 1 & 1 & 12 \\ 0 & 8 & 2 & 2 & -12 \\ 0 & 2 & 8 & 2 & 24 \\ 0 & 2 & 2 & 8 & -12 \end{array}\right) \overset{\substack{R_2 \to 1/2R_2 \\ R_3 \to 1/2R_3 \\ R_4 \to 1/2R_4 \\ R_2 \leftrightarrow R_3}}{\rightsquigarrow}$$

$$\left(\begin{array}{cccc|c} 3 & 1 & 1 & 1 & 12 \\ 0 & 1 & 4 & 1 & 12 \\ 0 & 4 & 1 & 1 & -6 \\ 0 & 1 & 1 & 4 & -6 \end{array}\right) \overset{\substack{R_3 \to 4R_2 - R_3 \\ R_4 \to R_2 - R_4}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 3 & 1 & 1 & 1 & 12 \\ 0 & 1 & 4 & 1 & 12 \\ 0 & 0 & 15 & 3 & 54 \\ 0 & 0 & 3 & -3 & 18 \end{array}\right) \overset{R_4 \to R_3 - 5R_4}{\rightsquigarrow}$$

$$\left(\begin{array}{cccc|c} 3 & 1 & 1 & 1 & 12 \\ 0 & 1 & 4 & 1 & 12 \\ 0 & 0 & 15 & 3 & 54 \\ 0 & 0 & 0 & 18 & -36 \end{array}\right)$$

Every variable being a pivot variable, there is a unique solution. By substitution, we find:

$u = -2$,

$15z - 6 = 54$, hence $z = 4$,

$y = -4z - u + 12 = -2$

$3x - 2 + 4 - 2 = 12$, hence $x = 4$.

3. (a) Let us denote by x_i the number of bikes in zone i in a stable distribution. If the number of bikes in each zone is the same at the beginning and end of each day, we have the following system of linear equation:

$$\begin{cases} 0.5x_1 + 0.4x_2 + 0.3x_3 + 0.1x_4 &= x_1 \\ 0.2x_1 + 0.2x_2 + 0.2x_3 + 0.2x_4 &= x_2 \\ 0.2x_1 + 0.2x_2 + 0.4x_3 + 0.2x_4 &= x_3 \\ 0.1x_1 + 0.2x_2 + 0.1x_3 + 0.5x_4 &= x_4 \end{cases}$$

This can be rewritten as

$$\begin{cases} -0.5x_1 + 0.4x_2 + 0.3x_3 + 0.1x_4 &= 0 \\ 0.2x_1 - 0.8x_2 + 0.2x_3 + 0.2x_4 &= 0 \\ 0.2x_1 + 0.2x_2 - 0.6x_3 + 0.2x_4 &= 0 \\ 0.1x_1 + 0.2x_2 + 0.1x_3 - 0.5x_4 &= 0 \end{cases}$$

or again

$$\begin{cases} -5x_1 + 4x_2 + 3x_3 + x_4 &= 0 \\ x_1 - 4x_2 + x_3 + x_4 &= 0 \\ x_1 + x_2 - 3x_3 + x_4 &= 0 \\ x_1 + 2x_2 + x_3 - 5x_4 &= 0 \end{cases}$$

We rearrange the equations and apply Gaussian elimination:

$$\left(\begin{array}{cccc|c} 1 & -4 & 1 & 1 & 0 \\ 1 & 1 & -3 & 1 & 0 \\ 1 & 2 & 1 & -5 & 0 \\ -5 & 4 & 3 & 1 & 0 \end{array}\right) \overset{\substack{R_2\to R_2-R_1 \\ R_3\to R_3-R_1 \\ R_4\to R_4+5R_1}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 1 & -4 & 1 & 1 & 0 \\ 0 & 5 & -4 & 0 & 0 \\ 0 & 6 & 0 & -6 & 0 \\ 0 & -16 & 8 & 6 & 0 \end{array}\right) \overset{\substack{R_3\to 1/6R_3 \\ R_4\to 1/2R_4 \\ R_2\leftrightarrow R_3}}{\rightsquigarrow}$$

$$\left(\begin{array}{cccc|c} 1 & -4 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 5 & -4 & 0 & 0 \\ 0 & -8 & 4 & 3 & 0 \end{array}\right) \overset{\substack{R_3\to R_3-5R_2 \\ R_4\to R_4+8R_2}}{\rightsquigarrow} \left(\begin{array}{cccc|c} 1 & -4 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & -4 & 5 & 0 \\ 0 & 0 & 4 & -5 & 0 \end{array}\right) \overset{R_4\to R_4+R_3}{\rightsquigarrow}$$

$$\left(\begin{array}{cccc|c} 1 & -4 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & -4 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The system is thus consistent, with free variable x_4. We set $x_4 = \alpha$ and find by substitution:

$x_4 = \alpha$,

$x_3 = \frac{5}{4}\alpha$,

$x_2 = \alpha$,

$x_1 = 4\alpha - \frac{5}{4}\alpha - \alpha = \frac{7}{4}\alpha$.

Moreover, as there are 200 bikes in total, we must have

$$200 = x_1 + \ldots + x_4 = 5\alpha$$

hence $\alpha = 40$, which yields:

$$(x_1, x_2, x_3, x_4) = (70, 40, 50, 40).$$

(b) The system of linear equations now becomes:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 &= x_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 &= x_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 &= x_3 \\ a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 &= x_4 \end{cases}$$

and the associated matrix is:

$$\begin{pmatrix} a_{11} - 1 & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} - 1 & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} - 1 & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} - 1 \end{pmatrix}$$

Now for every j, we have $a_{1j} + a_{2j} + a_{3j} + a_{4j} = 1$. This encodes the fact that, for every j, bikes parked at the beginning of the day in zone j will end up either in zone 1, 2, 3, or 4 (and with probability 1). We thus get

$$R_1 + R_2 + R_3 + R_4 = \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}.$$

When doing Gaussian elimination, we can thus start by the row operation $R_4 \rightsquigarrow R_1 + R_2 + R_3 + R_4$ (which is the concatenation of the three row operations $R_4 \rightsquigarrow R_3 + R_4$, $R_4 \rightsquigarrow R_2 + R_4$, and $R_4 \rightsquigarrow R_1 + R_4$). This kills the fourth row, and we then proceed by row operations only involving the first three rows. Once the (non-augmented) matrix is in row echelon form, we thus get a 4×4 matrix whose last row is zero. There are thus at most three rows contributing a pivot, hence at least one free variable. Thus, the system has infinitely many solutions by a result from the course.

Remark: In the final chapter of this course, we will see why there is always a stable distribution for such systems.

4. • We have $P(1) = a + b + c = 3, \ldots$. We thus have to solve the following system of linear equations:

$$\begin{cases} a + b + c &= 3 \\ 4a + 2b + c &= 3 \\ 9a + 3b + c &= 5 \\ a - b + c &= 6 \end{cases}$$

We apply Gaussian elimination

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 4 & 2 & 1 & 3 \\ 9 & 3 & 1 & 5 \\ 1 & -1 & 1 & 6 \end{array}\right) \overset{\substack{R_2 \to 4R_1 - R_2 \\ R_3 \to 9R_1 - R_3 \\ R_4 \to R_4 - R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 9 \\ 0 & 6 & 8 & 22 \\ 0 & -2 & 0 & 3 \end{array}\right) \overset{\substack{R_3 \to 3R_2 - R_3 \\ R_4 \to R_4 + R_2}}{\rightsquigarrow}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 9 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 3 & 12 \end{array}\right) \overset{R_4 \to 3R_3 - R_4}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right)$$

We see from the bottom line that the system is inconsistent, hence such a polynomial function does not exist.

• The equations translate into the following system:

$$\begin{cases} a - b + c &= 1 \\ a + b + c &= -1 \\ 2a + b &= 3 \\ 4a + 2b + c &= 4 \end{cases}$$

We apply Gaussian elimination

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ 2 & 1 & 0 & 3 \\ 4 & 2 & 1 & 4 \end{array}\right) \overset{\substack{R_2\to R_2-R_1\\ R_3\to R_3-2R_1\\ R_4\to R_4-4R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 0 & -2 \\ 0 & 3 & -2 & 1 \\ 0 & 6 & -3 & 0 \end{array}\right) \overset{\substack{R_3\to 3R_2-2R_3\\ R_4\to R_4-3R_2}}{\rightsquigarrow}$$
$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 0 & -2 \\ 0 & 0 & 4 & -8 \\ 0 & 0 & -3 & 6 \end{array}\right) \overset{R_4\to 4R_4+3R_3}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 0 & -2 \\ 0 & 0 & 4 & -8 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The system is consistent and has a unique solution as every variable is a pivot variable. By substitution, we then find: $c = -2$, $b = -1$, $a = 1$. Hence, $P(x) = 2x^2 - x - 2$.

5. (a) Let $(u_1, \ldots, u_n)^T$ and $(v_1, \ldots, v_n)^T$ be two solutions, and let $\lambda \in \mathbb{R}$. We have the following equation (E_1)

$$a_1u_1 + \cdots + a_nu_n = b$$

and the following equation (E_2):

$$a_1v_1 + \cdots + a_nv_n = b.$$

By performing $\lambda(E_1) + (1-\lambda)(E_2)$, we get:

$$\lambda(a_1u_1 + \cdots + a_nu_n) + (1-\lambda)(a_1v_1 + \cdots + a_nv_n) = \lambda b + (1-\lambda)b,$$

which can be rewritten as

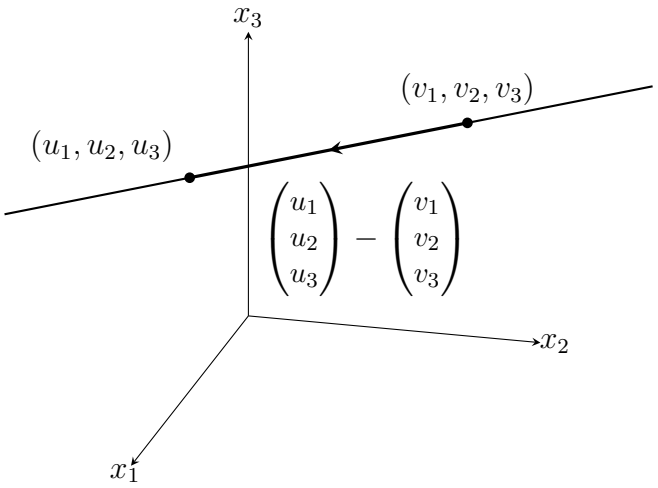
$$a_1(\lambda u_1 + (1-\lambda)v_1) + \cdots + a_n(\lambda u_n + (1-\lambda)v_n) = b.$$

In other words, $(\lambda u_1 + (1-\lambda)v_1, \ldots, \lambda u_n + (1-\lambda)v_n)^T$ is also a solution of the original equation.

Geometrically, if we think of $(u_1, \ldots, u_n)^T$ and $(v_1, \ldots, v_n)^T$ as points of $\mathbb{R}^n$, then the set of points of the form $(\lambda u_1 + (1-\lambda)v_1, \ldots, \lambda u_n + (1-\lambda)v_n)^T$, for $\lambda \in \mathbb{R}$, is the unique line of $\mathbb{R}^n$ going through those two points. In particular, it is clear that

$$\begin{pmatrix} \lambda u_1 + (1-\lambda)v_1 \\ \lambda u_2 + (1-\lambda)v_2 \\ \vdots \\ \lambda u_n + (1-\lambda)v_n \end{pmatrix} = \underbrace{\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}}_{\text{fixed point}} + \lambda \underbrace{\left[\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} - \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \right]}_{\text{direction vector}}$$

which resembles the vector equation of a line. A visualisation when $n = 3$ is as follows:



Note that the extra thick line corresponds to the line segment formed by the set of points where $0 \leq \lambda \leq 1$.

Thus, the solution set of a system of a linear equation contains *every line* going through any two distinct solutions.

(b) Question (a) dealt with a single equation, but the exact same reasoning holds for an arbitrary system: If $(u_1, \ldots, u_n)^T$ and $(v_1, \ldots, v_n)^T$ are two distinct solutions of this system, then for every $\lambda \in \mathbb{R}$,

$$(\lambda u_1 + (1-\lambda)v_1, \ldots, \lambda u_n + (1-\lambda)v_n)^T$$

is also a solution.

Suppose that the system has at least two distinct solutions $(u_1, \ldots, u_n)^T$ and $(v_1, \ldots, v_n)^T$. Then by (a), the system contains infinitely many solutions, namely all the solutions of the form $(\lambda u_1 + (1-\lambda)v_1, \ldots, \lambda u_n + (1-\lambda)v_n)^T$. Thus, this system can have, zero, one, or infinitely many solutions.

(c) • The system

$$\begin{cases} x &= 0 \\ y &= 0 \end{cases}$$

is a system whose solution set is a line, namely the z-axis.

• (This whole paragraph is much easier to understand if you draw a picture!) The union of two distinct lines Λ_1 and Λ_2 of $\mathbb{R}^3$ cannot be the solution set of a system of linear equations. Indeed, take a point $M_1 \in \Lambda_1$ that is not Λ_2, and a point $M_2 \in \Lambda_2$ that is not in Λ_1. Then the middle of the segment $[M_1M_2]$ is not in $\Lambda_1 \cup \Lambda_2$ (Why?), so in particular the line (M_1M_2) is not contained in $\Lambda_1 \cup \Lambda_2$. If $\Lambda_1 \cup \Lambda_2$ was a solution set, then by (a) it would in particular contain (M_1M_2).

• The system

$$\{ x = 0$$

is a plane of $\mathbb{R}^3$, namely the plane Oyz.

6. • We apply Gaussian elimination:

$$\left(\begin{array}{rrr|l} 2 & -1 & 3 & a \\ 3 & 1 & -5 & b \\ -5 & -5 & 21 & c \end{array}\right) \overset{\substack{R_2\to 2R_2-3R_1 \\ R_3\to 2R_3+5R_1}}{\rightsquigarrow} \left(\begin{array}{rrr|r} 2 & -1 & 3 & a \\ 0 & 5 & -19 & 2b-3a \\ 0 & -15 & 57 & 2c+5a \end{array}\right) \overset{R_3\to R_3+3R_2}{\rightsquigarrow}$$

$$\left(\begin{array}{rrr|r} 2 & -1 & 3 & a \\ 0 & 5 & -19 & 2b-3a \\ 0 & 0 & 0 & -4a+6b+2c \end{array}\right)$$

Row 3 corresponds to the equation $0x_1 + 0x_2 + 0x_3 = -4a + 6b + 2c$ so the system is inconsistent if and only if $-4a + 6b + 2c \neq 0$ i.e. $2a - 3b - c \neq 0$.

• We apply Gaussian elimination:

$$\left(\begin{array}{rrr|l} 1 & 1 & -1 & 2 \\ 1 & 2 & 1 & 3 \\ 1 & 1 & a^2-5 & a \end{array}\right) \overset{\substack{R_2\to R_2-R_1 \\ R_3\to R_3-R_1}}{\rightsquigarrow} \left(\begin{array}{rrr|r} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & a^2-4 & a-2 \end{array}\right)$$

The last row corresponds to the equation $(a^2-4)x_3 = a-2$. If $a^2-4 \neq 0$, i.e. $a \neq \pm 2$, every variable is a pivot variable and the system has a unique solution.
If $a = 2$ the last row becomes zero and the system has infinitely many solutions.
If $a = -2$ the last row corresponds to $0x_1+0x_2+0x_3 = -4$ so the system is inconsistent.